

02/28/29

CHE 321 HW 4

1) Q5-12

(a) NO, you cannot treat the volumetric flow rate as constant, but you can still substitute for it correctly if you express it as a function of conversion using the gas-law relationship. The issue is not substitution itself, the issue is assuming $v = v_0$. Total molar flow and pressure both change along the packed bed.

(b) Gas phase PBRs require coupling the mole balance with pressure drop.

$$\frac{dx}{dW} = \frac{-r_A}{F_{A0}}, \quad \frac{dP}{dW} = \text{Ergun equation}, \quad \text{conversion and pressure influence each other}$$

Reaction rate depends on conversion indirectly through concentration

$$\text{gas-phase reactors: } -r_A = K C_A^n, \quad C_A \neq C_{A0}(1-x)$$

$$C_A = C_{A0} \frac{(1-x) P}{(1+EX) P_0}$$

conversion affects rates through density changes, not just stoichiometry

(c) An inert dilutes the reactant, lowering C_A , which decreases reaction rate and requires a larger reactor (more catalyst weight) to achieve the same conversion

P5-1)

(b) The slope of p vs. z increases the most when the flow rate/velocity is increased, because the dominant (turbulent) Ergun term scales with velocity squared. The next strongest losses are decreasing particle diameter and decreasing porosity.

▷ both of which strongly increase resistance to flow

(vi) Pressure drop is a hydrodynamics / design constraint, not just a nuisance. As P falls down the bed, gas density decreases, which can raise volumetric flow (and velocities), further affecting ΔP and reactor performance

▷ laminar/viscous dominated: $\Delta P \sim \mu v_s / d_p^2$

▷ Turbulent/inertial dominated: $\Delta P \sim \rho v_s^2 / d_p$

(c) (i) K , C_{AO} , S_{AO} push X up; v_o pushes X down, & pushes P down and therefore also pushes X down via reduced concentrations

(ii) Because pressure drop becomes so severe near the end of the bed that the total concentration collapses

$$C_T = \frac{P}{RT} \Rightarrow C_T \downarrow \text{ as } P \downarrow$$

(iii) Ignoring pressure drop can badly overpredict conversion

▷ Pressure drops couples transport to kinetics via concentration

▷ Downstream catalyst can become "wasted"

▷ hevers that increase concentration (higher P_o , higher y_{AO} , higher C_{AO}) are powerful.

(e) (i) When K increases $\rightarrow X(w)$ increases, $P(w)$ is almost unchanged, $S(w)$ increases slightly

▷ when K decreases $\rightarrow X(w)$ decreases, $P(w)$ is unchanged, $S(w)$ changes only mildly, mostly still dominated by $1/P$

(i) When d increases $\rightarrow P(w)$ decreases faster and to a smaller exit value, $X(w)$ decreases, $f(w)$ increases significantly

When d decreases $\rightarrow P(w)$ stays higher, $X(w)$ increases, $f(w)$ is smaller

(ii) Pressure drop penalizes conversion.
Volumetric flow rises as pressure drops

(iii) If you are constrained to a maximum allowable pressure drop (say $\Delta P/P_0 \leq 0.3$), what changes to reactor design / operation would you consider first, and how would each affect x and required catalyst weight?

In a real packed bed, d depends on flow rate, gas properties, and pellet size (Ergun). How would increasing throughput change d , and what would that imply of the complex behavior of $P(w)$, $f(w)$ and $X(w)$

P5-5) PFR: $F_{A0} \frac{dx}{dV} = -r_A$

$$\Rightarrow V = F_{A0} \int_0^x \frac{dx}{-r_A}$$

$$-r_A = k C_A^2, \quad k = A e^{-\frac{E}{RT}}$$

Gas phase, must account of the change in total number of moles which affects v

$$\delta = \frac{1}{2} - 1 = -\frac{1}{2}, \quad y_{A(0)} = 1$$

$$\epsilon = (-\frac{1}{2})(1) = -\frac{1}{2}$$

$$C_A = C_{A0} \frac{(1-x)}{(1+\epsilon x)} \quad \text{or} \quad C_A = \frac{F_A}{V} = \frac{F_{A0} (1-x)}{V_0 (1+\epsilon x) \left(\frac{P_0}{P}\right) \left(\frac{T}{T_0}\right)}$$

Assumptions:

- no pressure drop $P_0 = P$
- isothermal, $T_0 = T$

$$\Rightarrow C_A = \frac{F_{A0} (1-x)}{(1 - \nu_2 x)}$$

$$\Rightarrow -r_A = k \left[\frac{C_{A0} (1-x)}{(1 - \nu_2 x)} \right]^2$$

$$\frac{dx}{dV} = \frac{k C_{A0}^2 (1-x)^2}{F_{A0} (1 - \nu_2 x)^2}$$

$$\text{Since } F_{A0} = V_0 C_{A0}, \quad \frac{dx}{dV} = \frac{k C_{A0}}{V_0} \frac{(1-x)^2}{(1 - \nu_2 x)^2}$$

For irreversible reaction, x never goes through a maximum
 x always increases with $\uparrow$ temp or $\uparrow$ volume,
 maximum only in reversible exothermic reacts.

x increases with increase Temp & volume; won't go through max because reaction is irreversible

P5-8) $-r_A = k C_A$; $\epsilon = 0$ (number of moles don't change)
 so $V = V_0$ & $C_A = C_{A0} (1-x)$

$$V = F_{A0} \int_0^x \frac{dx}{-r_A}$$

$A \rightarrow B$, $-r_A = k C_A$

$$V = F_{A0} \int_0^{0.632} \frac{dx}{-K C_A} = F_{A0} \int_0^{0.632} \frac{dx}{-K (C_{A0}(1-x))}$$

$$F_{A0} = V_0 C_{A0}$$

$$V = \frac{V_0}{K} \int_0^{0.632} \frac{dx}{(1-x)}$$

$$\frac{KV}{V_0} = [-\ln |1-x|]_0^{0.632} = -\ln(1-0.632) - (-\ln(1-0))$$

$$\Rightarrow T_K - \ln(0.368) = 1$$

$$\underline{T_A = 1}$$

$$V = \frac{V_0 C_{A0} x_1}{K C_{A0}(1-x)}$$

$$\frac{V \cdot K}{V_0} = T_K = \frac{x_1}{1-x_1} = 1 \Rightarrow 1 = x_1 = 1$$

$$\Rightarrow \text{So } x_1 = x_2$$

$$\Rightarrow 1 = \ln |1-x| \Big|_{x_2}^{x_2} = [-\ln |1-x_2| - (-\ln |1-x_2|)]$$

$$1 = \ln \left(\frac{x_2}{1-x_2} \right)$$

$$\Rightarrow x_2 = 0.816$$